$$ax^3 + bx^2 + cx + d = a(x-\alpha)(x-\beta)(x-\gamma)$$

Expanding the right hand side,

$$ax^3 - a(\alpha+\beta+\gamma)x^2 + a(\alpha\beta+\beta\gamma+\gamma\alpha)x - a(\alpha\beta\gamma).$$

Comparing the coefficients of like powers, we obtain

$$\alpha+\beta+\gamma = \frac{-b}{a},\ \alpha\beta+\beta\gamma+\gamma\alpha = \frac{c}{a} \text{ and } \alpha\beta\gamma = \frac{-d}{a}.$$

Since the degree of the polynomial equation is 3, we have $a \neq 0$ and hence division by $a$ is meaningful. If a monic cubic polynomial has roots $\alpha$, $\beta$, and $\gamma$, then

$$\text{coefficient of } x^2 = -(\alpha+\beta+\gamma),$$

$$\text{coefficient of } x = \alpha\beta+\beta\gamma+\gamma\alpha, \text{ and}$$

$$\text{constant term} = -\alpha\beta\gamma.$$

### (ii) Vieta's Formula for Polynomial equation of degree $n > 3$

The same is true for higher degree monic polynomial equations as well. If a monic polynomial equation of degree $n$ has roots $\alpha_1, \alpha_2, ..., \alpha_n$, then

| | | | |
|---|---|---|---|
| coefficient of $x^{n-1}$ | $= \sum_1$ | $=$ | $-\sum\alpha_1$ |
| coefficient of $x^{n-2}$ | $= \sum_2$ | $=$ | $\sum\alpha_1\alpha_2$ |
| coefficient of $x^{n-3}$ | $= \sum_3$ | $=$ | $-\sum\alpha_1\alpha_2\alpha_3$ |
| coefficient of $x$ | $= \sum_{n-1}$ | $=$ | $(-1)^{n-1}\sum\alpha_1\alpha_2...\alpha_{n-1}$ |
| coefficient of $x^0$ = constant term | $= \sum_n$ | $=$ | $(-1)^n\alpha_1\alpha_2...\alpha_n$ |

where $\sum\alpha_1$ denotes the sum of all roots, $\sum\alpha_1\alpha_2$ denotes the sum of product of all roots taken two at a time, $\sum\alpha_1\alpha_2\alpha_3$ denotes the sum of product of all roots taken three at a time, and so on. If $\alpha, \beta, \gamma,$ and $\delta$ are the roots of a quartic equation, then $\sum\alpha_1$ is written as $\sum\alpha$, $\sum\alpha_1\alpha_2$ is written as $\sum\alpha\beta$ and so on. Thus we have,

$$\begin{aligned}
\sum\alpha &= \alpha+\beta+\gamma+\delta \\
\sum\alpha\beta &= \alpha\beta+\alpha\gamma+\alpha\delta+\beta\gamma+\beta\delta+\gamma\delta \\
\sum\alpha\beta\gamma &= \alpha\beta\gamma+\alpha\beta\delta+\alpha\gamma\delta+\beta\gamma\delta \\
\sum\alpha\beta\gamma\delta &= \alpha\beta\gamma\delta
\end{aligned}$$



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